

A NOTE ON THE DIFFERENTIAL GEOMETRY OF LEAST SQUARES ESTIMATION FOR NONLINEAR REGRESSION MODELS

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Key words: Least squares; multiple stationary points; nonlinear models; sum of squares function.

Summary: The sum of squares function of a nonlinear regression model is itself nonlinear in the unknown parameters and may therefore have more than one stationary point with respect to those parameters. This situation leads to complications of a computational and an inferential nature and is therefore of some interest. In this article the relevant theory for characterizing observations on a nonlinear regression model for which the sum of squares function has more than one stationary point is developed within a differential-geometric framework.

1. INTRODUCTION

By definition the least squares estimators of the parameters of a regression model are those values of the parameters for which the residual sum of squares is a minimum. For nonlinear regression the sum of squares function is itself nonlinear in the parameters of the model and may well possess local as well as global minima. As a consequence least squares estimation is not straightforward. In particular the minimization of the sum of squares function involves nonlinear optimization and in general heuristic algorithms must be used to obtain solutions. Such algorithms may well display poor convergence properties and in any case do not necessarily guarantee that the global minimum may be found. Furthermore even if the global minimum of the sum of squares function can be located and the true least squares

estimator obtained, the problems of inference for the unknown parameters and of the sampling properties of the estimators still remain. For instance confidence regions for the parameters which are based on the sum of squares function may well be disjoint, while the sampling distribution of the least squares estimators may be skew or even bimodal. It is therefore of some interest to consider the occurrence or otherwise of multiple stationary points for the sum of squares function of nonlinear regression models.

Such considerations have been reported recently by Ross, both in his book (1990) and in a related article (1992). In particular Ross (1990, 1992) has observed that sample space can be partitioned into subspaces within which the number and nature of the stationary points of the sum of squares function are the same and has provided examples of nonlinear regression models to illustrate this observation. He has also presented techniques for identifying the occurrence of multiple optima for the sum of squares function in certain cases. Other than this, however, limited attention has been given in the literature to actually identifying and examining situations in which the sum of squares function for a nonlinear regression model has more than one stationary point, although in this context the studies of Williams (1962), Ross (1975, 1982), Clarke (1982) and Pázman (1984, 1987) and the comments of Beale (1960) should be mentioned.

The present article is *expository* in nature, and its aims are two-fold.

- (1) To draw together results in differential geometry which pertain to the existence of multiple optima for the sum of squares function of a nonlinear regression model, and which are scattered throughout the mathematics literature, and in so doing to make these results more readily available to the practitioner.
- (2) To provide a general framework for appraising such multiple optima as they occur in practice and, more especially, in the course of simulations.

2. MULTI-PARAMETER MODELS

Consider the nonlinear regression model in the form

$$(2.1) \quad y^i = \eta(\mathbf{x}^i, \boldsymbol{\theta}) + \epsilon^i, \quad i = 1, \dots, n$$

where y^i is an observation taken at the i -th experimental setting, $\mathbf{x}^i$, $\boldsymbol{\theta} = (\theta^1, \dots, \theta^p)$ is a $p \times 1$ vector of unknown parameters in some parameter space, $\boldsymbol{\theta}$, η is a known nonlinear response function which is twice continuously differentiable in $\boldsymbol{\theta}$, and the errors, ϵ^i , are uncorrelated with mean zero and constant variance, σ^2 . For convenience, the expected response, $\eta(\mathbf{x}^i, \boldsymbol{\theta})$, is written as $\eta^i(\boldsymbol{\theta})$ or, more concisely, as η^i and also the summation convention for repeated subscripts and superscripts is used throughout this article. The sum of squares of residuals for the model (2.1) is given by

$$S(\boldsymbol{\theta}; \mathbf{y}) = (y^i - \eta^i(\boldsymbol{\theta}))\delta_{ij}(y^j - \eta^j(\boldsymbol{\theta}))$$

where $\mathbf{y} = (y^1, \dots, y^n)$ and δ_{ij} is 1 for $i = j$ and zero otherwise, and the least squares estimator of $\boldsymbol{\theta}$, denoted $\hat{\boldsymbol{\theta}}$, is that value of $\boldsymbol{\theta}$ at which $S(\boldsymbol{\theta}; \mathbf{y})$ has a global minimum for $\boldsymbol{\theta} \in \Theta$.

As $\boldsymbol{\theta}$ changes over parameter space, Θ , so the vector of expected responses, $\boldsymbol{\eta} = (\eta^1, \dots, \eta^n)$, generates a p -dimensional curved surface, S_p , in the sample space, $\mathbb{R}^n$. This surface is termed the solution locus, or expectation surface, and inherits the fundamental metric tensor

$$g_{rs} = \eta_r^i \delta_{ij} \eta_s^j,$$

where $\eta_r^i = \frac{\partial \eta^i}{\partial \theta^r}$ is a covariant tensor with respect to $\boldsymbol{\theta}$ -transformations,

from the enveloping Euclidean space, $\mathbb{R}^n$ (McCullagh and Cox, 1986). Note that the form of S_p depends on the response function, η , and on the choice of design points, $\mathbf{x}^i$, in model (2.1) but is independent of the parameterization of the model.

2.1 Stationary points of the sum of squares, $S(\theta, y)$

Consider now an observation, y , and suppose that the corresponding sum of squares function, $S(\theta; y)$, has a stationary point at θ^* . Then θ^* is a solution to the p equations, $\frac{\partial S(\theta; y)}{\partial \theta} = 0$, and hence to the least squares equations

$$(y^i - \eta^i) \delta_{ij} \eta_r^j = 0, \quad r = 1, \dots, p.$$

The latter equations can be interpreted geometrically as follows. The p vectors with components, $\eta_r^i(\theta^*)$, span the space tangential to the solution locus, S_p , at θ^* . Thus the vector with components $y^i - \eta^i(\theta^*)$ is normal to S_p and can be expressed as

$$(2.2) \quad y^i - \eta^i(\theta^*) = \omega N^i,$$

where $N = (N^1, \dots, N^n)$ is the unit vector normal to S_p at θ^* in the same direction as $y - \eta(\theta^*)$, and $\omega > 0$ is the length of $y - \eta(\theta^*)$.

The nature of the stationary point of $S(\theta; y)$ at θ^* can be determined from the definiteness or otherwise of the Hessian matrix, $W(\theta) = \left[\frac{\partial^2 S(\theta; y)}{\partial \theta^r \partial \theta^s} \right]$, evaluated at θ^* . To be specific, if $W(\theta^*)$ is positive definite then $S(\theta; y)$ has a local minimum at θ^* , if $W(\theta^*)$ is negative definite then $S(\theta; y)$ has a local maximum at θ^* and otherwise $S(\theta; y)$ has a saddle point at θ^* . Now the rs -th element of $W(\theta)$ is given by

$$W_{rs}(\theta) = -2(y^i - \eta^i) \delta_{ij} \eta_{rs}^j + 2 \eta_r^i \delta_{ij} \eta_s^j$$

where $\eta_{rs}^j = \frac{\partial^2 \eta^j}{\partial \theta^r \partial \theta^s}$ and is not a tensor. Let b_{rs} be the second fundamental tensor of the surface, S_p , at θ^* viewed from the direction, N , and defined by $b_{rs} = N^i \delta_{ij} \eta_{rs}^j$, where η_{rs}^j is the tensor or covariant derivative of η_r^j . Then, since $\eta_{rs}^j = \eta_{rs}^j - \{rs\}_i \eta_i^j$, where $\{rs\}_i$ is the Christoffel symbol

for S_p (McCullagh and Cox, 1986) and $N^i \delta_{ij} \eta_t^j = 0$, it follows that $b_{rs} = N^i \delta_{ij} \eta_{rs}^j$. Thus, for $\theta = \theta^*$, the expression (2.2) holds and

$$(y^i - \eta^i) \delta_{ij} \eta_{rs}^j = \omega N^i \delta_{ij} \eta_{rs}^j = \omega b_{rs}$$

so that $W_{rs}(\theta^*) = -2\omega b_{rs} + 2g_{rs}$, or, in matrix form

$$(2.3) \quad W(\theta^*) = -2\omega [b_{rs}] + 2[g_{rs}]$$

where the matrices $[g_{rs}]$ and $[b_{rs}]$ are evaluated at θ^* . Now $[g_{rs}]$ is positive definite for all values of θ , and hence it follows from (2.3), and by invoking a little matrix algebra, that $W(\theta^*)$ is positive definite if and only if the latent roots of the matrix, $M(\theta^*) = [\delta_s^r - \omega g^{ru} b_{us}]$, are strictly positive, and $W(\theta^*)$ is negative definite if and only if these roots are all strictly negative. The latent roots of $M(\theta^*)$ are readily derived as follows. The principal curvatures of S_p at θ^* viewed from the direction N , denoted $\gamma^1, \gamma^2, \dots, \gamma^p$, and with maximum and minimum values γ_{max} and γ_{min} respectively, satisfy the determinantal equation

$$|b_{rs} - \gamma g_{rs}| = 0$$

(McCullagh and Cox, 1986), and are thus the latent roots of the matrix, $[g^{ru} b_{us}]$. Hence the latent roots of $M(\theta^*) = [\delta_s^r - \omega g^{ru} b_{us}]$ are given by $1 - \omega \gamma^1, 1 - \omega \gamma^2, \dots, 1 - \omega \gamma^p$. Overall therefore, if $1 - \omega \gamma^r > 0$ for all $r = 1, \dots, p$, then $W(\theta^*)$ is positive definite and $S(\theta; y)$ has a minimum at θ^* , if $1 - \omega \gamma^r < 0$ for all $r = 1, \dots, p$ then $W(\theta^*)$ is negative definite and $S(\theta; y)$ has a maximum at θ^* , and otherwise $S(\theta; y)$ has a saddle point at θ^* .

More particularly, suppose that the principal curvatures of S_p at θ^* viewed from the direction N , viz. γ^r , $r = 1, \dots, p$, are strictly positive. Then, since ω is positive by definition, it follows that $S(\theta; y)$ has a local

minimum at θ^* if $\omega < \frac{1}{\gamma_{\max}}$, a saddle point at θ^* if $\frac{1}{\gamma_{\max}} \leq \omega \leq \frac{1}{\gamma_{\min}}$, and a local maximum at θ^* if $\omega > \frac{1}{\gamma_{\min}}$. Similarly if the principal curva-

tures, γ^r , $r = 1, \dots, p$, are all strictly negative, then the terms, $1 - \omega\gamma^r$, $r = 1, \dots, p$, are all strictly positive, and $S(\theta; y)$ has a local minimum at θ^* . Finally if certain of the principal curvatures, γ^r , $r = 1, \dots, p$, are positive, and others are negative, then $S(\theta; y)$ has a minimum at θ^* if $\omega \leq \frac{1}{\gamma_{\max}}$, and a saddle point at θ^* otherwise; in this case $S(\theta; y)$ has no maxima. Overall therefore the points on the normal to S_p at θ^* with coordinates,

$$(2.4) \quad y^i = \eta^i(\theta^*) + \frac{1}{\gamma^r} N^i, \quad r = 1, \dots, p,$$

define the boundaries in sample space at which the nature of the stationary point of $S(\theta; y)$ at θ^* changes.

These results are illustrated for 3-dimensional sample space and a 2-dimensional solution locus, S_2 , in figures 1(a) and 1(b). There is clearly only one normal to S_2 at any point on that surface. The vectors, N_1 and N_2 , represent unit normals to S_2 at θ^* with origin, $\eta(\theta^*)$, and with opposite directions, and the points, G_1 and G_2 , have co-ordinates satisfying equation (2.4), and correspond to centres of principal curvature (Goetz, 1970, p. 303). The point, $\eta(\theta^*)$, on S_2 shown in figure 1(a) is elliptic, and the principal curvatures, γ^1 and γ^2 , viewed from the direction, N_1 , are both positive, with $\gamma^1 > \gamma^2$. Thus for an observation, y , such that $y - \eta(\theta^*) = \omega N_1$ with $\omega > 0$, the nature of the stationary point of $S(\theta; y)$ at θ^* is determined by the size of ω , as illustrated in figure 1(a). In contrast, the curvatures, γ^1 and γ^2 , viewed from the direction, N_2 , are both negative, and thus for a point, y , such that $y - \eta(\theta^*) = \omega N_2$ with $\omega > 0$, $S(\theta; y)$ has a minimum at θ^* irrespective of the magnitude of ω . In figure 1(b), $\eta(\theta^*)$ is a hyperbolic point, and the principal curvatures, γ^1

and γ^2 , viewed from either of the directions, N_1 and N_2 , have opposite signs. It thus follows that for any observation, y , lying on the normal to S_2 at θ^* , the nature of the stationary point of $S(\theta; y)$ at θ^* is determined by the size of ω , as shown in figure 1(b).

Finally it should be noted that Pázman (1987) considered points in sample space for which there is exactly one solution to the least squares equations, and showed, by invoking arguments similar to those used here, that such a solution corresponds to a global minimum.

2.2 Intersection of normals to the solution locus, S_p

Consider now observations, y , for which the sum of squares function, $S(\theta; y)$, has more than one stationary point. In such cases, the least squares equations have multiple solutions, and thus there exists more than one normal from y to the solution locus, S_p . Clearly the existence of such multiple normals is a global property of the entire surface, S_p , embedded in $\mathbb{R}^n$, and is thus dependent on the specific model being considered. However some insight into locating those points in sample space for which $S(\theta; y)$ has more than one stationary point can be gained by considering the local intersection of normals to S_p .

In particular consider a point on the normal, N , to S_p at θ^* with coordinates

$$z^i = \eta^i(\theta^*) + \rho N^i,$$

and let N represent the space normal to S_p at θ^* . Suppose that the vector, θ^* , undergoes an infinitesimal displacement, $d\theta^* = (d\theta^1, \dots, d\theta^p)$, in parameter space, Θ . Then the corresponding normal to S_p at $\theta^* + d\theta^*$ is defined by $z^i + dz^i$ where

$$(2.5) \quad dz^i = d\eta^i + \rho dN^i + d\rho N^i,$$

and, following the terminology used by Weatherburn (1927, section 29) for differential geometry in 3-dimensions, such a normal is referred to in this article as a normal consecutive to N^i and hence consecutive to the normal space, $\mathcal{N}$. Now the normal defined by $z^i + dz^i$ will intersect the space $\mathcal{N}$ if and only if the displacement dz^i occurs in that space, i.e. $dz^i \delta_{ij} \eta_s^j = 0$. But from (2.5),

$$dz^i = \eta_r^i d\theta^r + \rho N_r^i d\theta^r + d\rho N^i,$$

where $N_r^i = \frac{\partial N^i}{\partial \theta^r}$, so that

$$dz^i \delta_{ij} \eta_s^j = \eta_r^i \delta_{ij} \eta_s^j d\theta^r + \rho N_r^i \delta_{ij} \eta_s^j d\theta^r + d\rho N^i \delta_{ij} \eta_s^j.$$

Further $N^i \delta_{ij} \eta_s^j = 0$, and also differentiating this latter equation with respect to θ^r gives $N_r^i \delta_{ij} \eta_s^j = -N^i \delta_{ij} \eta_{rs}^j = -b_{rs}^i$, so that

$$dz^i \delta_{ij} \eta_s^j = (g_{rs} - \rho b_{rs}^i) d\theta^r.$$

Thus $dz^i \delta_{ij} \eta_s^j = 0$ if and only if $(g_{rs} - \rho b_{rs}^i) d\theta^r = 0$, and thus if and only if $\rho = \frac{1}{\gamma^t}$, $t = 1, \dots, p$, where γ^t are the principal curvatures of S_p at θ^* viewed from the direction N , with attendant principal directions defined by the appropriate differentials $(d\theta^1, \dots, d\theta^p)$. Overall therefore normals consecutive to the normal space, $\mathcal{N}$, will intersect that space at the points on the normal, N^i , to S_p at θ^* defined by

$$(2.6) \quad z^i = \eta^i(\theta^*) + \frac{1}{\gamma^t} N^i, \quad t = 1, \dots, p.$$

The above approach follows closely that of Eisenhart (1926, section 57) and is a generalization of that given by Weatherburn (1957, section 87) for hypersurfaces.

2.3 Partitioning hypersurfaces

The results of sections 2.1 and 2.2 can now be drawn together as follows.

Firstly observe that a point on the normal to S_p at θ^* with coordinates defined by (2.4), or equivalently (2.6), is a point at which (i) the nature of the stationary point of $S(\theta; y)$ at θ^* changes and (ii) normals consecutive to the space, $\mathcal{N}$, normal to S_p at θ^* intersect that space. Now suppose that the normal, N , to S_p at θ^* is allowed to vary over the space, $\mathcal{N}$; then for a given index, r , the point defined by (2.4) generates an $(n-p-1)$ -dimensional surface in $\mathcal{N}$. Further, as θ^* is allowed to vary over parameter space, Θ , so the points of this $(n-p-1)$ -dimensional surface in turn generate an $(n-1)$ -dimensional surface, i.e. a hypersurface, H_r , in $\mathbb{R}^n$. Overall therefore there are p such hypersurfaces, H_r , $r = 1, \dots, p$, corresponding to the p principal curvatures, γ_r^* , $r = 1, \dots, p$, and these hypersurfaces partition the sample space, $\mathbb{R}^n$, into subspaces such that the number and nature of the stationary points of $S(\theta; y)$ corresponding to observations, y , within a given subspace are the same.

These results are illustrated for the Michaelis-Menten model,

$$(2.7) \quad y^i = \frac{\theta^1 x^i}{\theta^2 + x^i} + \epsilon^i, \quad i = 1, 2, 3$$

with $\theta^1, \theta^2 > 0$ in figure 2. In particular, sample space is 3-dimensional, the solution locus, S_2 , is a ruled surface generated by the θ^2 -coordinate lines, and there is one surface, H_1 , defined by (2.4), which is the envelope of the normals to S_2 . The surfaces, S_2 and H_1 , for the design, $x = (0.5; 2.6904; 5)$, are shown in figure 2, viewed down the y^3 -axis from the plane, $y^3 = 5$, through the planes $y^3 = 3.75$, $y^3 = 2.5$, and $y^3 = 1.25$, to the origin, and selected θ^2 -coordinate lines for both surfaces are also included in that figure. Further algebraic details relating to the equations defining these surfaces are given in the appendix.

3. ONE-PARAMETER MODELS

The results of section 2 hold for one-parameter nonlinear regression models, but with some interesting modifications. In particular, the solution locus is a curve, C , in sample space, $\mathbb{R}^n$, and the fundamental metric tensor can be expressed as

$$(3.1) \quad \left[\frac{ds}{d\theta} \right]^2 = \frac{d\eta^i}{d\theta} \delta_{ij} \frac{d\eta^j}{d\theta},$$

where s is the arc-length parameter for C . Then $\frac{d\eta}{ds}$ is the unit tangent vector to C at s , and the vector $\frac{d^2\eta}{ds^2}$, termed the first curvature vector, is normal to the tangent vector and can be written as

$$(3.2) \quad \frac{d^2\eta}{ds^2} = \kappa \mathbf{n},$$

where κ is the first curvature of C at s with $\kappa > 0$, and $\mathbf{n} = (n^1, \dots, n^n)$ the corresponding principal normal vector (Weatherburn, 1957, section 42).

Consider now an observation, y . Then the sum of squares function, $S(\theta; y)$, has a stationary point at θ^* if and only if the vector with coordinates, $y^i - \eta^i(\theta^*)$, is normal to the curve C at θ^* , and thus if and only if θ^* is a solution to the least squares equation

$$(3.3) \quad (y^i - \eta^i) \delta_{ij} \frac{d\eta^j}{d\theta} = 0.$$

The nature of this stationary point is determined by the sign of $W(\theta^*)$, where

$$W(\theta) = \frac{d^2 S(\theta; y)}{d\theta^2} = -2(y^i - \eta^i) \delta_{ij} \frac{d^2 \eta^j}{d\theta^2} + 2 \frac{d\eta^i}{d\theta} \delta_{ij} \frac{d\eta^j}{d\theta}.$$

Further, by observing that $\frac{d^2 \eta^j}{d\theta^2} = \left[\frac{ds}{d\theta} \right]^2 \frac{d^2 \eta^j}{ds^2} + \left(\frac{d^2 s}{d\theta^2} \right) \frac{d\eta^j}{ds}$, and by expressing $y^i - \eta^i(\theta^*)$ as ωN^i , where $\omega > 0$ and $N = (N^1, \dots, N^n)$ is a unit vector normal to C at θ^* , it then follows from (3.1), (3.2) and (3.3) that

$$W(\theta^*) = 2(1 - \omega \kappa N^i \delta_{ij} n^j) \left(\frac{ds}{d\theta} \right)^2 \Big|_{\theta=\theta^*}.$$

Thus $W(\theta^*)$ is equivalent in sign to $1 - \omega \kappa N^i \delta_{ij} n^j$, and hence to $1 - \omega \gamma$, where $\gamma = \kappa N^i \delta_{ij} n^j$ is the first curvature of C at θ^* viewed down the normal, N , and to $1 - \lambda \kappa$ where $\lambda = \omega N^i \delta_{ij} n^j$ is the scalar projection of $y - \eta(\theta^*)$ onto n . Thus for $N^i \delta_{ij} n^j > 0$, it follows that if $\omega < \frac{1}{\gamma}$ or equivalently $\lambda < \frac{1}{\kappa}$, then $W(\theta^*)$ is positive and hence $S(\theta; y)$ has a minimum at θ^* , and that otherwise $W(\theta^*)$ is negative and $S(\theta; y)$ has a maximum at θ^* . In contrast, for $N^i \delta_{ij} n^j < 0$, $W(\theta^*)$ is always positive, and thus $S(\theta; y)$ has a minimum at θ^* . These results are illustrated for a curve, C , in 3-dimensional sample space in figure 3. The plane, M , normal to C at θ^* is shown in that figure, together with the principal normal vector, n , the centre of curvature, G , and the binormal vector, b (Goetz, 1970, sections 5 and 6). In this case, $N^i \delta_{ij} n^j = \cos \alpha$, where α is the angle subtended by the vectors, N and n in M , and thus $\lambda = \omega \cos \alpha$ and $\gamma = \kappa \cos \alpha$. For all observations, y , falling on the plane, M , with $\lambda < \frac{1}{\kappa}$, $S(\theta; y)$ has a minimum at θ^* , and for all other observations on that plane, $S(\theta; y)$ has a maximum, as indicated in figure 3.

Also for one-parameter nonlinear regression models the intersection of normals to C can be examined in a more direct manner than that used for the multi-parameter case. Thus the hyperplanes normal to C defined by the least squares equation (3.3) constitute a one-parameter family so that, following Kreyszig (1968, p. 253), the intersection of the normal hyperplane to C at θ^* with that to C at $\theta^* + h$, in the limit as h approaches zero, is given by the $(n-2)$ -dimensional flat

$$(3.4) \quad \begin{cases} (y^i - \eta^i) \delta_{ij} \frac{d\eta^j}{d\theta} = 0 \\ (y^i - \eta^i) \delta_{ij} \frac{d^2\eta^j}{d\theta^2} - \frac{d\eta^i}{d\theta} \delta_{ij} \frac{d\eta^j}{d\theta} = 0 \end{cases}$$

or equivalently by the set of points on the hyperplane normal to C at θ^* satisfying $(y^i - \eta^i)\delta_{ij}\pi^j = \lambda = \frac{1}{\kappa}$. Overall therefore it is clear that as θ varies over parameter space, Θ , so the flat (3.4) generates a hypersurface, H , which partitions sample space into regions within which the number and nature of the stationary points of $S(\theta; y)$ are the same. These results are illustrated for the exponential model,

$$y^i = e^{-\theta x^i} + \epsilon^i, \quad i = 1, 2,$$

with $\theta > 0$ and $x = (1, 10)$ in figure 4. Sample space is 2-dimensional, the solution locus, C , is described explicitly by $\eta^2 = (\eta^1)^{2/2.1}$, and the surface, H , defined by (3.4), is the evolute of the curve, C (Goetz, 1970, p. 66). For observations, y , falling into the regions labelled A_1 in figure 4, there is only one normal from y to C , and thus $S(\theta; y)$ has a unique minimum at θ , while for observations falling into the region labelled A_2 , there are three such normals to C , corresponding to two local minima and one maximum of $S(\theta; y)$.

4. CONCLUSIONS

The present study has shown that the partitioning of sample space for a nonlinear regression model into subspaces, within which the number and nature of the stationary points of the sum of squares function are the same, can be explained by considering the properties of the normals from points in that sample space to the solution locus. However, it is apparent that, for a given observation, it is not always possible to identify the particular subspace of sample space into which it falls, and thus to elucidate the number and nature of the stationary points of the corresponding sum of squares function. Indeed if this were possible, a broad problem in nonlinear optimization would be solved. It is also noteworthy that the differential-geometric approach presented here involves local properties of the solution locus, and that it is clearly not possible to accommodate the global properties thereof in a general way. For instance the solution locus of a nonlinear regression model is

usually bounded, and as a consequence observations for which there are no corresponding normals to the interior of the solution locus may occur, so that the least squares estimators then refer to collapsed forms of the original model. Overall therefore it may be concluded that many of the problems relating to the occurrence of multiple optima for the sum of squares function of a nonlinear regression are, to a greater extent, example-specific.

ACKNOWLEDGMENTS

The author would like to thank Professor G.P.Y. Clarke and Mrs J.M. Nevin for many helpful discussions, and a referee for some very constructive comments. This research was supported by grants from the Foundation for Research Development and from the University of Natal in South Africa.

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APPENDIX

The solution locus, S_2 , for the model, (2.7), is a ruled surface generated by the θ^2 -coordinate lines, and, more particularly, corresponds to a portion of a cone with vertex, 0. Its two boundary edges are defined by the coordinate line, $\theta^2 = 0$, and by the limiting values of the expected responses, η^1 , as θ^1 and θ^2 approach infinity with θ^1/θ^2 constant. The design, $\mathbf{z} = (0.5; 2.6904; 5)$, used for illustration in figure 2 was chosen to show the curvature of S_2 to best advantage. Also the θ^2 -coordinate lines for S_2 shown in that figure correspond to $\theta^2 = 0, \frac{1}{2}, 1, 2, 4, 8, 16$ and the limiting line as θ^2 approaches infinity. At any point on the solution locus, S_2 , there are two principal curvatures, one of which is zero. As a consequence there is only one surface, H_1 , defined by the set of points satisfying (2.4), and corresponding to the non-zero principal curvature, and this surface is defined explicitly as follows. The least squares equations for model (2.7) are given by

$$\begin{cases} \sum_{i=1}^3 \left(y^i - \frac{\theta^1 x^i}{\theta^2 + x^i} \right) \frac{x^i}{(\theta^2 + x^i)} = 0 \\ \sum_{i=1}^3 \left(y^i - \frac{\theta^1 x^i}{\theta^2 + x^i} \right) \frac{x^i}{(\theta^2 + x^i)^2} = 0, \end{cases}$$

and eliminating θ^1 from these equations gives the expression,

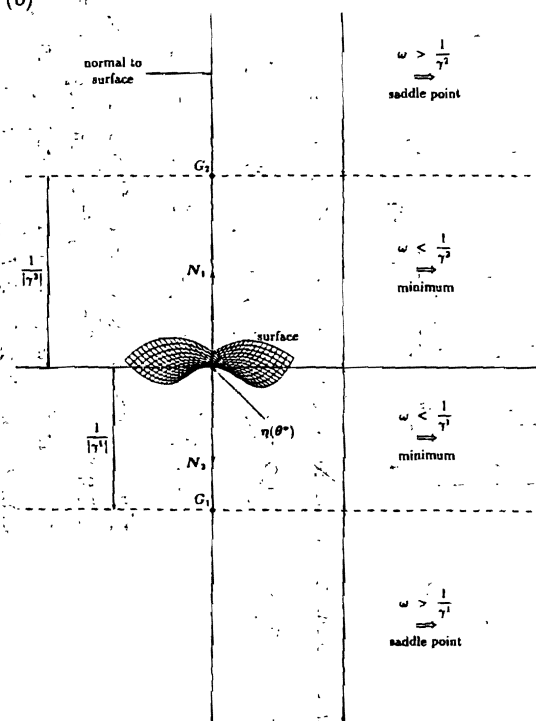
$$(A1) \quad \sum_i \sum_j \sum_{k \neq i,j} y^i x^j (x^j)^2 (x^i - x^j) \{ (\theta^2)^4 + (3x^k + x^i)(\theta^2)^3 + 3x^k(x^k + x^i)(\theta^2)^2 + (x^k)^2(x^k + 3x^i)(\theta^2) + x^i(x^k)^3 \} = 0 \quad \text{for } i, j, k = 1, 2, 3,$$

which defines the plane normal to S_2 along a θ^2 -coordinate line. It then follows from section 2.2 that the surface, H_1 , is the envelope of this family of normals, and hence, following Kreyszig (1968, p. 253), is defined by (A1), together with the equation obtained by differentiating (A1) with respect to θ^2 , viz.

$$(A2) \quad \sum_i \sum_j \sum_{k \neq i,j} y^i x^j (x^j)^2 (x^i - x^j) \{ 4(\theta^2)^3 + 3(3x^k + x^i)(\theta^2)^2 + 6x^k(x^k + x^i)(\theta^2) + (x^k)^2(x^k + 3x^i) \} = 0, \quad \text{for } i, j, k = 1, 2, 3.$$

Thus H_1 is also a ruled surface, with rulings corresponding to constant values of θ^2 . In addition, H_1 has an edge of regression which satisfies (A1), (A2), and the equation formed by differentiating (A1) twice with respect to θ^2 . This edge corresponds to a θ^2 value of 2.8915, and is shown in figure 2. Finally it should be noted that, in the context of 3-dimensional differential geometry, H_1 is termed the evolute of S_2 , or its focal or centre surface (see Kreyszig, 1968, p. 264; Goetz, 1970, p. 303).

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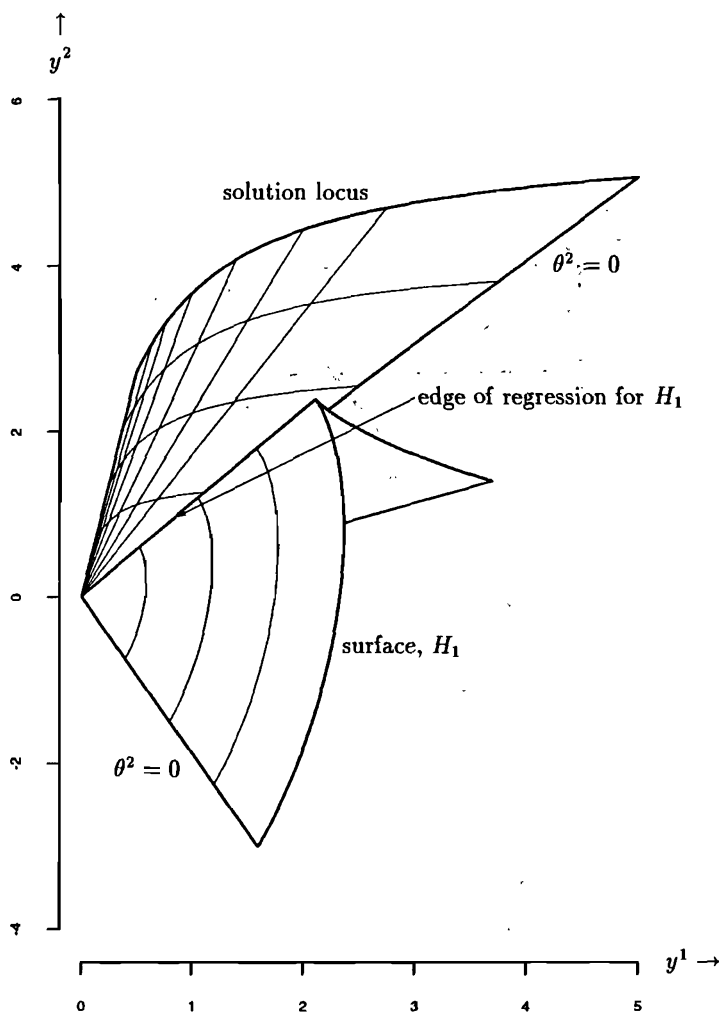


FIGURE 2. The solution locus, S_2 , and the surface, H_1 , in 3-dimensional sample space for the Michaelis-Menten model.

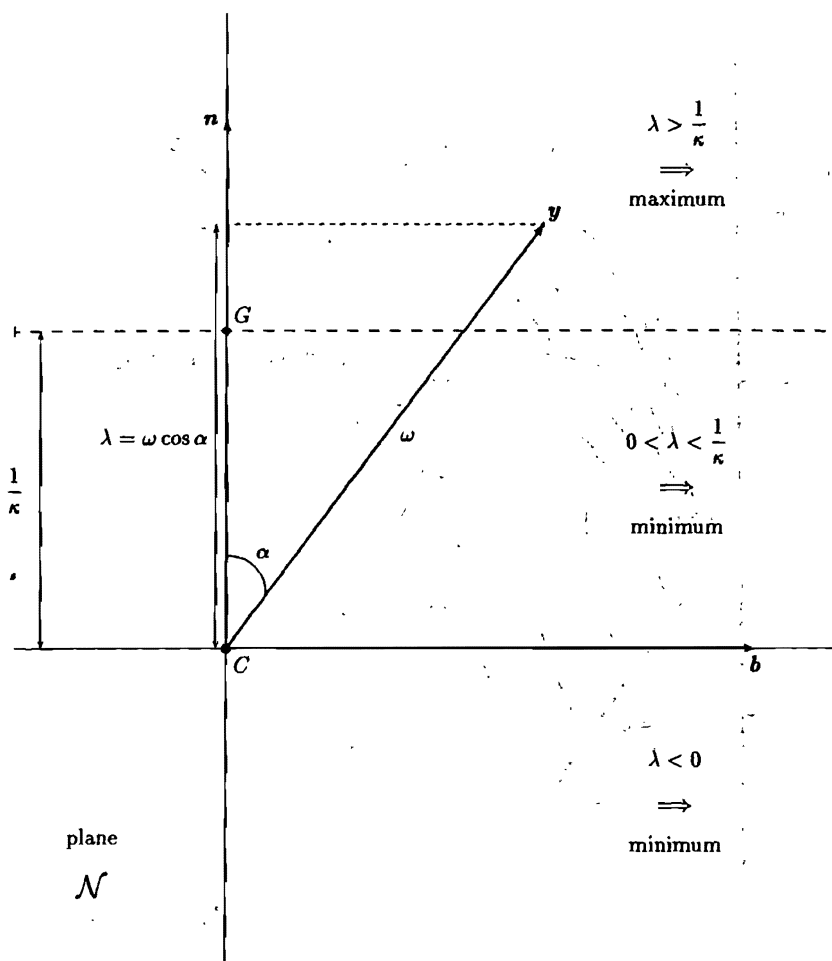


FIGURE 3. Diagram of the plane, $\mathcal{N}$, normal to a curve, C , in 3-dimensional sample space to illustrate the results of section 3.

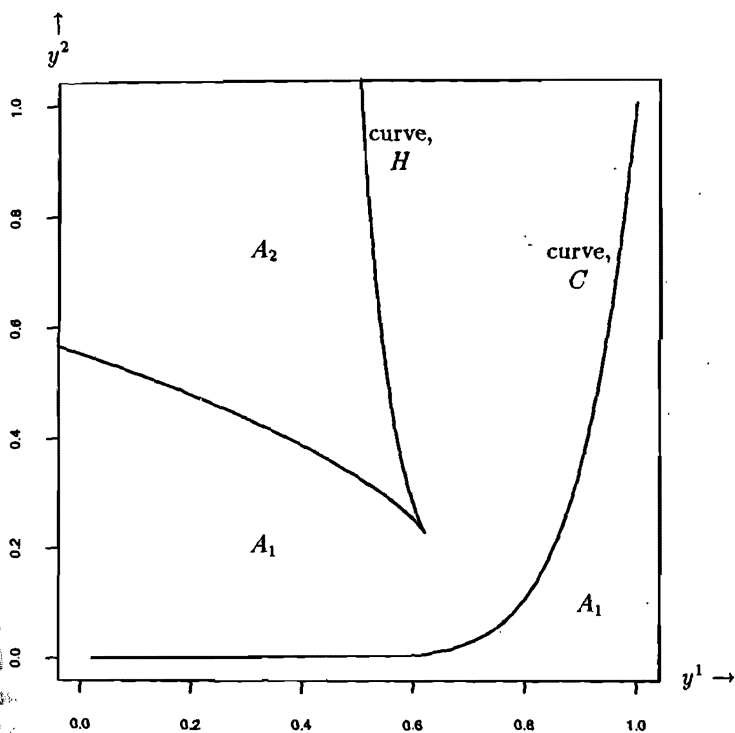


FIGURE 4. The solution locus, S , and the curve, H , in 2-dimensional sample space for the exponential model.